

Abrupt change in mean using block bootstrap and avoiding variance estimation

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Received: 24 February 2017 / Accepted: 8 December 2017 / Published online: 12 December 2017
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Abstract We deal with sequences of weakly dependent observations that are naturally ordered in time. Their constant mean is possibly subject to change at most once at some unknown time point. The aim is to test whether such an unknown change has occurred or not. The change point methods presented here rely on ratio type test statistics based on maxima of the cumulative sums. These detection procedures for the abrupt change in mean are also robustified by considering a general score function. The main advantage of the proposed approach is that the variance of the observations neither has to be known nor estimated. The asymptotic distribution of the test statistic under the no change null hypothesis is derived. Moreover, we prove the consistency of the test under the alternatives. A block bootstrap method is developed in order to obtain better approximations for the test's critical values. The validity of the bootstrap algorithm is shown. The results are illustrated through a simulation study, which demonstrates computational efficiency of the procedures. A practical application to real data is presented as well.

Institutional support to Barbora Peřtová was provided by RVO:67985807. The research of Michal Peřta was supported by the Czech Science Foundation Project GAČR No. 15-04774Y.

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Keywords Change point · Ratio type statistics · Hypothesis testing · Change in mean · Robustness · Block bootstrap

1 Introduction and main goals

The problem of an unknown change in mean of the time series is studied and procedures for detection of such change within the observed time ordered sequence are presented. The considered underlying stochastic model allows *at most one change*. Moreover, the change is a rapid—also called abrupt—change, which happens suddenly at some unknown time point.

The ratio type test statistic elaborated in this paper is derived from the non-ratio type test statistics (e.g., Csörgő and Horváth 1997) based on partial sums of the residuals that are commonly used in the change point analysis. They do not need to be standardized by any variance estimate, which makes them a suitable alternative for the non-ratio type test statistics, most of all in situations, when it is *difficult to find a variance estimate* with satisfactory properties. Such difficulty can occur in situations with dependent random errors. However, the variance estimators often do not perform well even in the i.i.d. case, especially under alternatives (Antoch et al. 1997).

We extend the ideas for the ratio type test statistics presented by Hušková (2007) and Horváth et al. (2008) in the way that *weakly dependent errors* of the model are considered together with incorporating a *general score function* in the test statistics. In order to obtain critical values for the studied test statistic not only from their asymptotic distributions, we focus on the *circular moving block bootstrap method* by Politis and Romano (1992). This type of resampling procedure was applied in a similar situation by Kirch (2006).

1.1 State of art and structure of the paper

Kim (2000) and, consequently, Kim and Amador (2002) studied how to detect a structural change characterized by a shift in persistence of linear time series using a similarly constructed ratio type test statistic. This work is further continued by the simulation study of Leybourne and Taylor (2006), where both asymptotic and finite sample properties of the proposed test are studied. In Hušková (2007), two ratio type test statistics based on the cumulative sums of the residuals are briefly introduced. These are intended to be used when testing whether the mean has changed at an unknown time and also when testing for a change from the asymptotic stationary sequence into the asymptotic difference stationary sequence. In Horváth et al. (2008), more details on the topic are given and other ratio type test statistics are introduced. The applicability of the method is demonstrated through a simulation study. Antoch and Hušková (2001), Hušková (2004), and Kirch (2006) dealt with permutation principles and bootstrapping in the change point analysis.

Generally, there are two main distinguishable classes of change point detection procedures: parametric (Csörgő and Horváth 1997, Chapter 1) and nonparametric (Csörgő and Horváth 1997, Chapter 2). The parametric ones rely on an underlying known parametric form of the distribution, where only the distributional parameters are unknown.

A typical representative of the parametric methods are the likelihood ratio (LR) based methods (Andrews 1993). If one assumes additional (regularity) assumptions, for instance, regarding the Fisher information matrix, some optimality properties can be derived. E.g., a Wald-like type or a Bahadur-like type of optimality. Optimal detections within the change point framework were investigated by, e.g., Andrews and Ploberger (1994) or Andrews et al. (1996). Although, one has to pay for these properties by assuming a specific parametric form of the distribution and by adding regularity conditions, which can be sometimes hard to verify. On the other hand, a classical representative of the nonparametric class are cumulative sums (CUSUM) based approaches that do not assume a parametric distribution, which gives them wider applicability. Our proposed methods belong to the CUSUM-based techniques.

Zhao et al. (2010, 2011) propose the ratio type tests to detect a change in the variance of linear processes. Furthermore, a new change point estimation method based on the ratio type statistics is introduced by Zhao et al. (2011). Robustification of the ratio type test statistic for change in mean detections was elaborated by Madurkayová (2011). Chen and Tian (2014) studied a ratio type test to detect the variance change in the nonparametric regression models under both fixed and random design cases. A similar ratio type statistic as introduced in this paper is studied to analyse change in mean for heavy tailed distributions in the work of Wang et al. (2016). This leads to a more general asymptotic distribution under the null hypothesis, which is a functional of stable Lévy processes. A bootstrap approximation method to determine the critical values and change point estimation using the ratio method are also discussed. Bazarova et al. (2015) develop a ratio type test to detect changes in the location parameters of dependent observations with infinite variances. The ratio type statistic based on the cumulative sums' process is adjusted by trimming off a set of observations that are the largest in magnitude. Moreover, the ratio and non-ratio type test statistics are compared by simulations. Peštová and Pešta (2015, 2016) applied the ratio type test statistics for detections of the structural changes in panel data.

The paper is structured as follows: Section 2 introduces an abrupt change point model together with stochastic assumptions. A ratio type test statistic for the change point detection is proposed in Sect. 3. Consequently, the asymptotic behavior of the considered test statistic is derived, which covers the first main theoretical contribution. A block bootstrap with replacement is shown in Sect. 4. Its justification is provided, which is the second main theoretical contribution. Section 5 contains a simulation study that compares performance of the asymptotic and bootstrap test. It numerically emphasizes the advantages and disadvantages of the proposed procedures. A practical application of the developed approach to an actuarial problem is presented in Sect. 6. Proofs are given in the “Appendix A”.

2 Abrupt change point model

Let us consider observations $Y_{1,n}, \dots, Y_{n,n}$ obtained at time ordered points. We are interested in testing the null hypothesis of all observations being random variables having equal mean values. Our goal is to test against the alternative of the first τ_n observations have mean value μ and the remaining $n - \tau_n$ observations come from

distributions with mean values $\mu + \delta_n$, $\delta_n \neq 0$. We can describe the situation as a model with *at most one abrupt change in constant mean*

$$Y_{k,n} = \mu + \delta_n \mathcal{I}\{k > \tau_n\} + \varepsilon_k, \quad k = 1, \dots, n, \quad (1)$$

where μ , δ_n and τ_n are unknown parameters, $\varepsilon_1, \dots, \varepsilon_n$ are random errors, and $\mathcal{I}\{A\}$ denotes the indicator of set A .

Despite the fact, that the observed data $\{Y_{k,n}\}_{k=1,n=1}^{n,\infty}$ form a stochastic *triangular array*, the random disturbances $\{\varepsilon_n\}_{n=1}^\infty$ are just a single sequence of random variables. So, the errors remain the same for each row of the triangular array of the observed variables. For the sake of convenience, we suppress the index n in the observations $Y_{k,n}$ as well as in the parameters δ_n and τ_n (and in the variables depending on the latter) whenever possible. However, we have to keep in mind that in the asymptotic results below, as n increases over all bounds, both $\delta = \delta_n$ and $\tau = \tau_n$ may be changing when n is increasing. Then, model (1) can be rewritten as the *abrupt change point model*

$$Y_k = \mu + \delta \mathcal{I}\{k > \tau\} + \varepsilon_k, \quad k = 1, \dots, n. \quad (\text{ACP})$$

The time point τ is called the *change point*.

We are going to test the null hypothesis that no change occurred

$$H_0 : \tau = n \quad (2)$$

against the alternative that a change occurred at some unknown time point τ

$$H_1 : \tau < n, \delta \neq 0. \quad (3)$$

3 Ratio type test statistic based on M -residuals

The main focus is given to the test procedures based on the ratio type test statistics that are functionals of the partial sums of M -type residuals. We robustify the original ratio type test statistic described by Huřková (2007) and Horváth et al. (2008). A test statistic based on the M -residuals is considered

$$\mathcal{R}_n(\psi, \gamma) = \max_{n\gamma \leq k \leq n-n\gamma} \frac{\max_{1 \leq i \leq k} \left| \sum_{j=1}^i \psi(Y_j - \hat{\mu}_{1k}(\psi)) \right|}{\max_{k \leq i \leq n-1} \left| \sum_{j=i+1}^n \psi(Y_j - \hat{\mu}_{2k}(\psi)) \right|}, \quad (4)$$

where $0 < \gamma < 1/2$ is a given constant, $\hat{\mu}_{1k}(\psi)$ is an M -estimate of parameter μ based on the observations $Y_1, \dots, Y_k$ (under H_0), and $\hat{\mu}_{2k}(\psi)$ is an M -estimate of μ based on the observations $Y_{k+1}, \dots, Y_n$ (under H_0). That means, $\hat{\mu}_{1k}(\psi)$ is a solution of the estimating equation

$$\sum_{i=1}^k \psi(Y_i - \mu) = 0$$

and, similarly, $\widehat{\mu}_{2k}(\psi)$ is a solution of the estimating equation

$$\sum_{i=k+1}^n \psi(Y_i - \mu) = 0.$$

Let us remark that the test statistic does not require an estimate of the possible change point. For the choice of $\psi_{L_2}(x) = x$, we get one of the statistics studied in Horváth et al. (2008). By considering different score functions, we may construct similar statistics, but more robust against outliers and more suitable for heavy tailed distributions.

3.1 Strong mixing dependence

Prior to deriving asymptotic properties of the test statistic, we summarize the notion of strong mixing (α -mixing) dependence in more detail, which will be imposed on the model's errors.

Suppose that $\{\varepsilon_n\}_{n=1}^\infty$ is a sequence of random elements on a probability space $(\Omega, \mathcal{F}, \mathbf{P})$. For sub- σ -fields $\mathcal{A}, \mathcal{B} \subseteq \mathcal{F}$, we define

$$\alpha(\mathcal{A}, \mathcal{B}) := \sup_{A \in \mathcal{A}, B \in \mathcal{B}} |\mathbf{P}(A \cap B) - \mathbf{P}(A)\mathbf{P}(B)|.$$

Intuitively, $\alpha(\cdot, \cdot)$ measures the dependence of the events in $\mathcal{B}$ on those in $\mathcal{A}$. There are many ways how to describe weak dependence or, in other words, *asymptotic independence* of random variables, see Bradley (2005). Considering a filtration $\mathcal{F}_m^n := \sigma\{\varepsilon_i \in \mathcal{F}, m \leq i \leq n\}$, sequence $\{\varepsilon_n\}_{n=1}^\infty$ of random variables is said to be *strong mixing* (α -mixing) if $\alpha(n) := \sup_{k \in \mathbb{N}} \alpha(\mathcal{F}_1^k, \mathcal{F}_{k+n}^\infty) \rightarrow 0$ as $n \rightarrow \infty$. Loosely speaking, the coefficients of dependence $\alpha(n)$ measure how much dependence exists between the events separated by at least n observations or time periods. Note that in case of a strictly stationary sequence, the $\sup_{k \in \mathbb{N}}$ in the definition becomes redundant.

Finite order processes, which do not satisfy Doeblin's condition, can be shown to be α -mixing (Ibragimov and Linnik 1971, pp. 312–313). Rosenblatt (1971) provides general conditions under which stationary Markov processes are α -mixing. Since functions of mixing processes are themselves mixing (Bradley 2005), time-varying functions of any of the above mentioned processes just mentioned are α -mixing as well.

3.2 Limit distribution under null hypothesis and alternative

We proceed to the assumptions that are needed for deriving asymptotic properties of the proposed test statistic. Before that, assumptions for the score function ψ and the distribution of the random errors $\varepsilon_1, \dots, \varepsilon_n$ are formulated.

Assumption A1 The random error terms $\{\varepsilon_i, i \in \mathbb{N}\}$ form a strictly stationary α -mixing sequence with marginal distribution function F , that is symmetric around zero, and for some $\chi > 0, \chi' > 0$ there exists a constant $C_1(\chi, \chi') > 0$ such that

$$\sum_{h=0}^{\infty} (h+1)^{\chi/2} \alpha(h)^{\chi'/(2+\chi+\chi')} \leq C_1(\chi, \chi'), \quad (5)$$

where $\alpha(k)$, $k = 0, 1, \dots$ are the α -mixing coefficients.

Assumption A2 The score function ψ is a non-decreasing and antisymmetric function.

Assumption A3 $\int |\psi(x)|^{2+\chi+\chi'} dF(x) < \infty$ and

$$\begin{aligned} \int |\psi(x+t_2) - \psi(x+t_1)|^{2+\chi+\chi'} dF(x) \\ \leq C_2(\chi, \chi') |t_2 - t_1|^\eta, |t_j| \leq C_3(\chi, \chi'), \quad j = 1, 2 \end{aligned}$$

for some constants $1 \leq \eta \leq 2 + \chi + \chi'$, $\chi > 0$, $\chi' > 0$ as in (5) and the constants $C_2(\chi, \chi') > 0$, $C_3(\chi, \chi') > 0$ both depending only on χ and χ' .

Assumption A4 Let us denote $\lambda(t) = -\int \psi(e-t) dF(e)$, for $t \in \mathbb{R}$. We assume that $\lambda(0) = 0$ and that there exists a first derivative $\lambda'(\cdot)$ that is Lipschitz in the neighborhood of 0 and satisfies $\lambda'(0) > 0$.

Assumption A5 $0 < \sigma^2(\psi) = \mathbb{E} \psi^2(\varepsilon_1) + 2 \sum_{i=1}^{\infty} \mathbb{E} \psi(\varepsilon_1) \psi(\varepsilon_{i+1}) < \infty$.

Assumption A1 is satisfied for example for the ARMA processes with continuously distributed stationary innovations and bounded variance (Doukhan 1994, Section 2.4). The conditions regarding ψ reduce to the moment restrictions for $\psi_{L_2}(x) = x$ (L_2 method) taking $\eta = 2 + \chi + \chi'$. For $\psi_{L_1}(x) = \text{sgn}(x)$ (L_1 method), the conditions reduce to F being a symmetric distribution, having continuous density f in a neighborhood of 0 with $f(0) > 0$, and $\eta = 1$ for any $\chi > 0$ and $\chi' > 0$. Similarly, we may consider the derivative of the Huber loss function, i.e.,

$$\psi_H(x) = x \mathbb{I}\{|x| \leq C\} + C \text{sgn}(x) \mathbb{I}\{|x| > C\} \quad (6)$$

for some $C > 0$. In that case to satisfy Assumptions A2–A4, we need to assume F being a symmetric distribution function with the continuous density f in a neighborhood of C and $-C$ satisfying $f(C) > 0$ and $f(-C) > 0$ with $\eta = 2 + \chi + \chi'$.

For $\psi_{L_2}(x) = x$, $x \in \mathbb{R}$, the above stated Assumptions A2 and A4 are satisfied. We can also drop the requirement of symmetry of F in Assumption A1 and replace it by $\mathbb{E} \varepsilon_1 = 0$. Assumptions A3 and A5 reduce to the following two assumptions.

Assumption A6 $\mathbb{E} |\varepsilon_1|^{2+\beta} < \infty$ for some constant $\beta > 0$.

Assumption A7 $0 < \sigma^2(\psi_{L_2}) = \mathbb{E} \varepsilon_1^2 + 2 \sum_{i=1}^{\infty} \mathbb{E} \varepsilon_1 \varepsilon_{i+1} < \infty$.

Although we consider only symmetric distribution function F in our approach, there is also another common source of outlying observations which occur in practical applications—asymmetric random error distribution (i.e., $\nexists x_0 \in \mathbb{R} : F(x_0 - x) =$

$1 - F(x_0 + x)$, $\forall x > 0$). To generalize our approach for asymmetric distribution function F , one needs to assure in modified Assumption A4 that there exists some unique $t_0 \in \mathbb{R}$ such that $\lambda(t_0) = 0$. Correspondingly, all other assumptions required to hold in the neighborhood of $t = 0$ need to be satisfied in the neighborhood of $t = t_0$ instead.

The following theorem states the asymptotic behavior of the studied ratio type test statistic under the null hypothesis.

Theorem 1 (Under null) *Suppose $Y_1, \dots, Y_n$ follow model (ACP) and assume that Assumptions A1–A5 hold. Then, under null hypothesis (2)*

$$\mathcal{R}_n(\psi, \gamma) \xrightarrow[n \rightarrow \infty]{\mathcal{D}} \sup_{\gamma \leq t \leq 1-\gamma} \frac{\sup_{0 \leq u \leq t} |\mathcal{W}(u) - u/t \mathcal{W}(t)|}{\sup_{t \leq u \leq 1} |\tilde{\mathcal{W}}(u) - (1-u)/(1-t) \tilde{\mathcal{W}}(t)|}, \quad (7)$$

where $\{\mathcal{W}(t), 0 \leq t \leq 1\}$ is a standard Wiener process and $\tilde{\mathcal{W}}(t) = \mathcal{W}(1) - \mathcal{W}(t)$.

The null hypothesis is rejected for large values of $\mathcal{R}_n(\psi, \gamma)$. We reject H_0 at significance level α if $\mathcal{R}_n(\psi, \gamma) > a_{1-\alpha, \gamma}$, where $a_{1-\alpha, \gamma}$ is the $(1 - \alpha)$ -quantile of the asymptotic distribution from (7). Explicit form of the limit distribution (7) under the null hypothesis is not known. Therefore, in order to obtain the critical values, we have to use either simulation from the limit distribution or resampling methods.

The lower bound for k in (4) may be relaxed to 1 (or 2, since the ratio is equal to 0 for $k = 1$) and, correspondingly, $\sup_{\gamma \leq t \leq 1-\gamma}$ in the limit distribution of (7) may be replaced with $\sup_{0 < t \leq 1-\gamma}$. However, this does not remain true for the supremum of the upper bound, i.e., $\sup_{\gamma \leq t \leq 1-\gamma}$ cannot be replaced by $\sup_{0 < t < 1}$ nor can it be replaced by $\sup_{\gamma \leq t < 1}$, since $\lim_{t \rightarrow 1-} \sup_{t \leq s \leq 1} |\mathcal{W}(s) - \frac{s}{t} \mathcal{W}(t)| = 0$ a.s. Nevertheless, one may redefine the test statistics $\mathcal{R}_n(\psi, \gamma)$ by interchanging its numerator and denominator in order to relax the upper bound for k up to n (or $n - 1$, since the reversed ratio becomes 0 for $k = n$) and, correspondingly, $\sup_{\gamma \leq t \leq 1-\gamma}$ in the new limit distribution may be replaced with $\sup_{\gamma < t \leq 1}$.

Next, we study the behavior of the test statistic under the alternative.

Theorem 2 (Under alternative) *Suppose $Y_1, \dots, Y_n$ follow model (ACP), assume that $\delta_n = O(n^\theta)$ as $n \rightarrow \infty$ for $\theta \in \left(-\frac{1}{2}, \frac{\eta}{3(2+\chi+\chi')} - \frac{1}{2}\right)$, and $\tau = [\zeta n]$ for some $\gamma < \zeta < 1 - \gamma$. Then, under Assumptions A1–A5 and alternative (3)*

$$\mathcal{R}_n(\psi, \gamma) \xrightarrow[n \rightarrow \infty]{\mathbb{P}} \infty.$$

Theorem 2 says that in presence of the structural change in mean, the test statistic explodes above all bounds. Hence, the procedure is consistent and the asymptotic distribution from Theorem 1 can be used to construct the test.

The asymptotic results for $\mathcal{R}_n(\psi_{L_2}, \gamma)$ were derived in Hušková (2007) and also in Horváth et al. (2008) under different assumptions regarding the random errors.

Table 1 Simulated critical values corresponding to the asymptotic distribution of the test statistic $\mathcal{R}_n(\psi, \gamma)$ under the null hypothesis

$100(1 - \alpha)\%$	90%	95%	97.5%	99%
$\gamma = 0.1$	6.298815	7.293031	8.283429	9.589896
$\gamma = 0.2$	4.117010	4.745884	5.368286	6.159252

For other score functions ψ , results regarding the limit behavior under fixed as well as under local alternatives for the related non-ratio type test statistic are presented in Huřková and Maruřiaková (2012). The result for the ratio type statistic under fixed alternative can be derived by a modification of the proof therein.

3.3 Asymptotic critical values

The explicit form of the limit distribution (7) is not known. The critical values may be determined by simulations from the limit distribution from Theorem 1. Theorem 2 ensures that we reject the null hypothesis for large values of the test statistic. We tried to simulate the asymptotic distribution (7) by *discretizing* the Wiener process and using the relationship of a random walk to the Wiener process. We considered 1000 as the number of discretization points within $[0, 1]$ interval and the number of simulation runs equals to 100,000. In Table 1, we present several critical values for $\gamma = 0.1$ and $\gamma = 0.2$.

Note that the numerator and denominator in the test statistic $\mathcal{R}_n(\psi, \gamma)$ can be interchanged and such a modified test statistic can still be used for detection of the abrupt change in mean (but using different critical values).

4 Block bootstrap with replacement

There are several different approaches that may be used when resampling dependent observations. Classical resampling methods are not suitable, since they do not take into account the underlying dependency structure. Here, we focus our attention on a so-called *circular moving block bootstrap method*, which was introduced by Politis and Romano (1992). Overlapping blocks of consequent observations are formed from the original observations. The first few consequent observations from the original sequence are appended after the last observation, so that for a sequence of length n , we always have n possible blocks of subsequent observations to choose from

$$\{(Y_{j+1}, \dots, Y_{j+K}), j = 0, \dots, n - 1\}; \quad \text{where } Y_i = Y_{i-n}, i > n.$$

With this method, there is equal probability for each observation to be included in the bootstrap sample.

Let L denote the number of blocks and let K be the block length. In order to keep the notation as simple as possible, we restrict ourselves to situation, where $n = KL$, i.e., a situation where the set of n observations can be divided in exactly L blocks of length K . It can be proved, as in Kirch (2006), that the limit results remain the same after omitting the last K_1 observations, if $n = KL + K_1$, $1 \leq K_1 \leq K - 1$. We will assume that K and n are both functions of L such that $n = KL$. Moreover, we will suppose that

$$L \rightarrow \infty \quad \text{and} \quad K \rightarrow \infty \quad \text{as} \quad n \rightarrow \infty.$$

It is also possible to use the (standard) non-circular moving block bootstrap (Künsch 1989), where one does not append the first few consequent observations from the original sequence after the last observation. This bootstrap version gives $n - K$ blocks to choose from (instead of n blocks), but we will not concentrate on this approach here.

First, let us define the following subsets of $\mathbb{N} \times \mathbb{N}$ for integer numbers l, k, L, K and real number $0 < \gamma < 1/2$

$$\begin{aligned} \Pi_{l,k,L,K} &= \{(p, q) : p, q \in \mathbb{N}, \\ &\quad 1 \leq p \leq l, 1 \leq q \leq K, (p-1)K + q \leq (l-1)K + k\}, \\ \tilde{\Pi}_{l,k,L,K} &= \{(p, q) : p, q \in \mathbb{N}, \\ &\quad l \leq p \leq L, 1 \leq q \leq K, (p-1)K + q \geq (l-1)K + k + 1\}, \\ \Omega_{L,K}(\gamma) &= \{(l, k) : l, k \in \mathbb{N}, \\ &\quad 1 \leq l \leq L, 1 \leq k \leq K, KL\gamma \leq (l-1)K + k \leq KL(1-\gamma)\}. \end{aligned}$$

For a set of i.i.d. random variables $\mathbf{U} = (U_1, \dots, U_L)$, uniformly distributed on the set $\{0, \dots, n-1\}$, we define the following block bootstrap statistic

$$\begin{aligned} S_{L,K}^U(p, q, l, k) &= \sum_{i=1}^{p-1} \sum_{j=1}^K \psi(Y_{U_i+j} - m_{L,K}^U(l, k)) \\ &\quad + \sum_{j=1}^q \psi(Y_{U_p+j} - m_{L,K}^U(l, k)), \end{aligned}$$

for $p, l = 1, \dots, L, q, k = 1, \dots, K, p \leq l, (p-1)K + q \leq (l-1)K + k$, where $m_{L,K}^U(l, k)$ is an M -estimate that solves the estimating equation

$$\sum_{r=1}^{l-1} \sum_{s=1}^K \psi(Y_{U_r+s} - \mu) + \sum_{s=1}^k \psi(Y_{U_l+s} - \mu) = 0$$

with respect to μ . Similarly, we define

$$\begin{aligned}\tilde{S}_{L,K}^U(p, q, l, k) &= \sum_{j=k+1}^K \left(Y_{U_l+j} - \tilde{m}_{L,K}^U(l, k) \right) \mathcal{I}\{p \geq l+1\} \\ &\quad + \sum_{i=l+1}^{p-1} \sum_{j=1}^K \left(Y_{U_i+j} - \tilde{m}_{L,K}^U(l, k) \right) \mathcal{I}\{p \geq l+2\} \\ &\quad + \sum_{j=1}^q \left(Y_{U_p+j} - \tilde{m}_{L,K}^U(l, k) \right),\end{aligned}$$

for $p, l = 1, \dots, L, q, k = 1, \dots, K$ such that $p \geq l, (p-1)K+q \geq (l-1)K+k+1$ and $\tilde{m}_{L,K}^U(l, k)$ is an M -estimate that solves the estimating equation

$$\sum_{s=k+1}^K \psi(Y_{U_l+s} - \mu) + \sum_{r=l+1}^L \sum_{s=1}^K \psi(Y_{U_r+s} - \mu) = 0.$$

Now, define the block bootstrap version of $\mathcal{R}_n(\psi)$ from (4) by

$$\mathcal{R}_{L,K}^*(\psi, \gamma) = \max_{(l,k) \in \Omega_{L,K}(\gamma)} \frac{\max_{(p,q) \in \Pi_{l,k,L,K}} |S_{L,K}^U(p, q, l, k)|}{\max_{(p,q) \in \tilde{\Pi}_{l,k,L,K}} |\tilde{S}_{L,K}^U(p, q, l, k)|}.$$

Statistic $\mathcal{R}_{L,K}^*(\psi, \gamma)$ is constructed in a similar fashion as the original ratio type test statistic $\mathcal{R}_n(\psi, \gamma)$. The idea behind the bootstrap test statistic lies in indexing the randomly chosen (possibly overlapping) bootstrap blocks by $l = 1, \dots, L$. The first l blocks are used in the numerator of the bootstrap statistic. The l th block is employed in the numerator as well as in the denominator. The last $L-l+1$ blocks are used in the denominator of the statistic $\mathcal{R}_{L,K}^*(\psi, \gamma)$. Regarding the l th block appearing in the numerator and denominator, this particular block is split into two continuous disjunctive parts: the first one contains the first elements from the l th block, has k elements, and is used for the numerator; the second part contains the last elements from the l th block, has $K-k$ elements, and is used for the denominator. So, there does not exist a bootstrapped observation appearing simultaneously in the numerator and denominator. An algorithm for the circular block bootstrap is illustratively shown in Procedure 4.1 and its validity will be proved in Theorem 3.

We are going to show that the bootstrapped ratio type test statistic, conditioned on the original observations, has exactly the *same limit behavior* as the original test statistic under the null. It does not matter whether our observations come from the null hypothesis or the alternative. In other words, we are going to prove that $\mathcal{R}_{L,K}^*(\psi, \gamma)$ provides asymptotically correct critical values for the test based on $\mathcal{R}_n(\psi, \gamma)$, when the observations follow either the null hypothesis or the alternative.

Theorem 3 (Bootstrap consistency) *Suppose $Y_1, \dots, Y_n$ follow model (ACP) and $\mathbb{E} |\psi(\varepsilon_1)|^v < \infty$ for some $v > 2$. Let Assumptions A1 and A3 be satisfied for $\chi_1, \chi'_1 >$*

Procedure 4.1 Bootstrapping test statistic $\mathcal{R}_n(\psi, \gamma)$ **Input:** Sequence of observations $Y_1, \dots, Y_n$, block length K , and $0 < \gamma < 1/2$.**Output:** Bootstrap distribution of $\mathcal{R}_n(\psi, \gamma)$, i.e., the empirical distribution where probability mass $1/B$ concentrates at each of $(1)\mathcal{R}_{L,K}^*(\psi, \gamma), \dots, (B)\mathcal{R}_{L,K}^*(\psi, \gamma)$.

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1: determine number of blocks  $L = \lceil n/K \rceil$ 
2: define set  $\Omega_{L,K}(\gamma)$ 
3: for  $b = 1$  to  $B$  do // repeat in order to obtain the empirical distribution
4:   generate random sample  $U = (U_1, \dots, U_L)$  from discrete uniform distribution on  $\{0, \dots, n-1\}$ 
5:   for  $(l, k) \in \Omega_{L,K}(\gamma)$  do
6:     define sets  $\Pi_{l,k,L,K}$  and  $\tilde{\Pi}_{l,k,L,K}$ 
7:     calculate  $(b)m_{L,K}^U(l, k)$  and  $(b)\tilde{m}_{L,K}^U(l, k)$ 
8:     for  $(p, q) \in \Pi_{l,k,L,K}$  do
9:       calculate  $(b)s_{L,K}^U(p, q, l, k)$ 
10:    end for
11:    compute  $\max_{(p,q) \in \Pi_{l,k,L,K}} |(b)s_{L,K}^U(p, q, l, k)|$ 
12:    for  $(p, q) \in \tilde{\Pi}_{l,k,L,K}$  do
13:      calculate  $(b)\tilde{s}_{L,K}^U(p, q, l, k)$ 
14:    end for
15:    compute  $\max_{(p,q) \in \tilde{\Pi}_{l,k,L,K}} |(b)\tilde{s}_{L,K}^U(p, q, l, k)|$ 
16:  end for
17:  compute bootstrap test statistics  $(b)\mathcal{R}_{L,K}^*(\psi, \gamma)$ 
18: end for

```

0 and for $\chi_2, \chi_2' > 0$ such that $\kappa < \chi_1 < v - 2$, $\chi_1' = v - 2 - \chi_1$, and $\chi_2 > 2 + 2\kappa$ for some $0 < \kappa < v - 2$. Moreover, let Assumptions A2, A4, and A5 be satisfied. Under alternative (3), let $\delta_n = O(n^\theta)$ as $n \rightarrow \infty$ for $\theta \in \left(-\frac{1}{2}, \frac{\eta}{3(2+\chi+\chi')} - \frac{1}{2}\right)$ and $\tau = \lfloor n\zeta \rfloor$ for some $\zeta : \gamma < \zeta < 1 - \gamma$. If $K = O(L)$ as $L \rightarrow \infty$, then we have for all $y \in \mathbb{R}$, as $L \rightarrow \infty$,

$$\begin{aligned} & \mathbf{P} \left[\mathcal{R}_{L,K}^*(\psi, \gamma) \leq y | Y_1, \dots, Y_n \right] \\ & \xrightarrow{\mathbf{P}} \mathbf{P} \left[\sup_{\gamma \leq t \leq 1-\gamma} \frac{\sup_{0 \leq u \leq t} |\mathcal{W}(u) - u/t\mathcal{W}(t)|}{\sup_{t \leq u \leq 1} |\tilde{\mathcal{W}}(u) - (1-u)/(1-t)\tilde{\mathcal{W}}(t)|} \leq y \right], \end{aligned}$$

where $\{\mathcal{W}(t), 0 \leq t \leq 1\}$ is a standard Wiener process and $\tilde{\mathcal{W}}(t) = \mathcal{W}(1) - \mathcal{W}(t)$.

The conditional weak convergence in probability from the above stated Theorem 3 can be replaced by the conditional weak convergence almost surely using Theorem 3.5.1, Remark 3.5.4 from Kirch (2006), and strengthening some assumptions.

A choice of the block length K in the circular moving block bootstrap is an important decision. It will affect the bootstrapped version of the test statistic. Therefore, the block length can be viewed as a tuning parameter in the circular moving block bootstrap procedure. One possibility, how to make such optimal choice, is to minimize the asymptotic mean square error of the circular moving block bootstrap variance estimate. Fitzenberger (1997) proved that this approach yields $K = O(n^{1/3})$ as $n \rightarrow \infty$ in case of the α -mixing random errors. In contrast to this theoretical asymptotic result, the practical choice of the block length usually needs to be made based on one finite

sample consisting of n observations. Several finite sample approaches for choosing the block length K were proposed by Hall et al. (1995), Politis and White (2004), and Lahiri et al. (2007).

5 Simulations

We were interested in the performance of the tests based on the ratio type test statistics $\mathcal{R}_n(\psi, \gamma)$ and $\mathcal{R}_{L,K}^*(\psi, \gamma)$ with $\psi_{L_2}(x) = x$ and $\psi_{L_1}(x) = \text{sgn}(x)$. We focused on the comparison of the accuracy of critical values obtained by the circular moving block bootstrap method with the accuracy of critical values obtained by the simulation from the limit distribution.

In Figs. 1 and 2, one may see *size-power plots* or size-power curves (SPC) for choices of $n = 200$, $\gamma = 0.1$, $\tau = n/2$, and $\delta = 1$ considering the test statistics $\mathcal{R}_n(\psi, \gamma)$ and $\mathcal{R}_{L,K}^*(\psi, \gamma)$ in case of the L_2 and L_1 score function both under the null hypothesis and under the alternative. The size-power plots illustrate the power of a test. The empirical distribution function of the p values of the test statistic for the null hypothesis (Fig. 1) or a given alternative (Fig. 2) is plotted with respect to the distribution used to determine the critical values of the test. What we get is a plot that shows the actual α -errors, respectively $1 - (\beta\text{-errors})$, on the y -axis for the chosen quantiles on the x -axis. The ideal situation under the null hypothesis is depicted by the straight diagonal dotted line. Under the alternative, the desired situation would be a steep function with values close to 1. For more details on the size-power plots we may refer, e.g., to Kirch (2006). The random errors were simulated as an AR(1) process with autoregression coefficients 0.3 (orange) and 0.5 (green), and as a set of i.i.d. random errors (blue). The rejection rates based on simulated asymptotic critical values are depicted by a dashed line, the rejection rates based on block bootstrap with block length $K = 5$ are depicted by a solid line. The standard normal distribution and the Student t -distribution with 5 degrees of freedom are used for generating the innovations of the models' errors.

We generated 10,000 independent samples in order to compute the asymptotic critical values. When bootstrapping, for each sample we used 1000 bootstrap samples to compute the bootstrap critical values. In the simulations of the rejection rates, we used 1000 repetitions.

In all of four subfigures of Fig. 1 depicting a situation under the null hypothesis, we may see that comparing to the critical values obtained by simulations from the asymptotic distribution, the critical values obtained by bootstrapping are more accurate, especially for the AR(1) sequences. When comparing the accuracy of α -errors for different choices of the score function ψ , the L_1 method seems to perform better than the L_2 method. However, when using the L_1 method, the power of the test slightly decreases (see Fig. 2).

With the choice of ψ_{L_2} , the simulated rejection rates under H_0 are higher than the corresponding theoretical α -levels for larger values of the autoregression coefficient, while for the L_1 method they remain much more stable. Comparing the case of $N(0, 1)$ innovations with the case of t_5 innovations, the rejection rates for the L_1 version of the

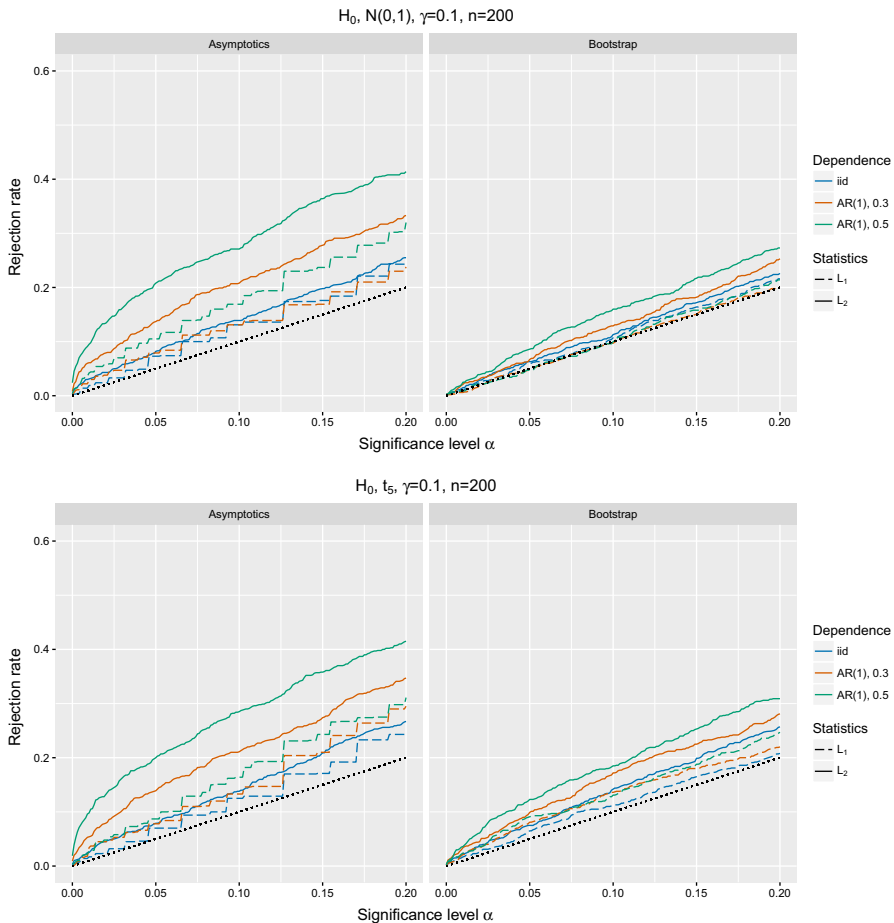


Fig. 1 Size-power plots for $\mathcal{R}_n(\psi, \gamma)$ and $\mathcal{R}_{L,K}^*(\psi, \gamma)$ under H_0 , where $\gamma = 0.1$

test statistic tend to be slightly higher for the t_5 distribution, while they remain more or less the same for the L_2 version.

Additionally, one can use a size-power plot with the *adjusted* (empirical) α -errors to compare the performance of $\mathcal{R}_n(\psi, \gamma)$ against $\mathcal{R}_{L,K}^*(\psi, \gamma)$. The *empirical size-power plots* in Fig. 3 display the empirical size of the test (i.e., $1 - \text{sensitivity}$) on the x -axis versus the empirical power of the test (i.e., specificity) on the y -axis. The ideal shape of the curve is as steep as possible. The empirical size-power plots demonstrate that the bootstrap ratio type test statistic $\mathcal{R}_{L,K}^*(\psi, \gamma, \cdot)$ gives approximately the same (only slightly smaller) empirical powers for the adjusted empirical sizes comparing to the original test statistic $\mathcal{R}_n(\psi, \gamma)$. This is due to two opposing facts: $\mathcal{R}_{L,K}^*(\psi, \gamma)$ keeps the significance level of the test better, but $\mathcal{R}_n(\psi, \gamma)$ gives higher power of the test.

As expected, the accuracy of the critical values tends to be better for larger n , which is demonstrated by a choice of $n = 500$ in Table 2. Furthermore, we possess some empirical results from our simulation study that are not presented in this paper: The

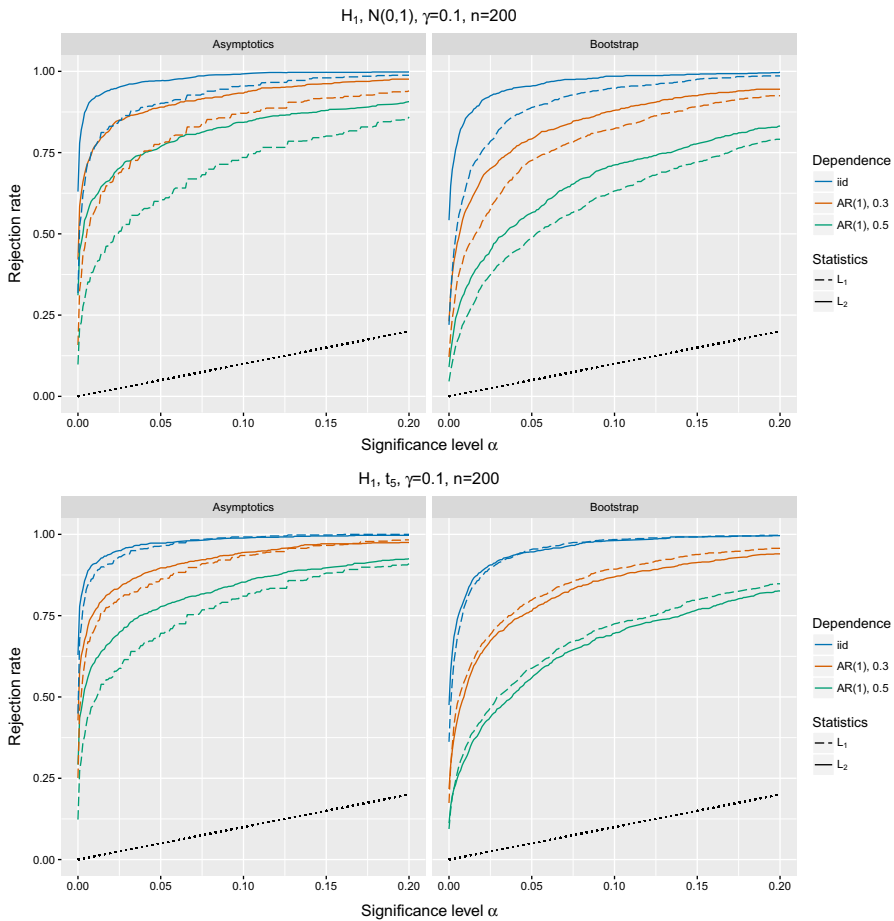


Fig. 2 Size-power plots for $\mathcal{R}_n(\psi, \gamma)$ and $\mathcal{R}_{L,K}^*(\psi, \gamma)$ under H_1 with $\tau = n/2$ and $\delta = 1$, where $\gamma = 0.1$

choice of $\gamma = 0.2$ seems to provide more accurate critical values than the choice of $\gamma = 0.1$, but the test power is larger in the latter case. We may also conclude that with larger abrupt change, the power of the test increases.

Finally, a comparison with standard and widely used change point detection procedures is provided with emphasis on computational performance. A classical representative from the *nonparametric* class of methods is a *cumulative sums (CUSUM)* test statistic

$$\mathcal{C}_n(M) := \frac{1}{\sqrt{\widehat{\sigma}_n^2(M)}} \max_{1 \leq k \leq n-1} \left| \sum_{i=1}^k (Y_i - \bar{Y}_n) \right| \quad (8)$$

where $\bar{Y}_n = 1/n \sum_{j=1}^n Y_j$ is the sample mean and $\widehat{\sigma}_n^2(M)$ is a suitable variance estimator. The null hypothesis is rejected for large values of $\mathcal{C}_n(M)$. For surveys, we refer to, e.g., Perron (2006).

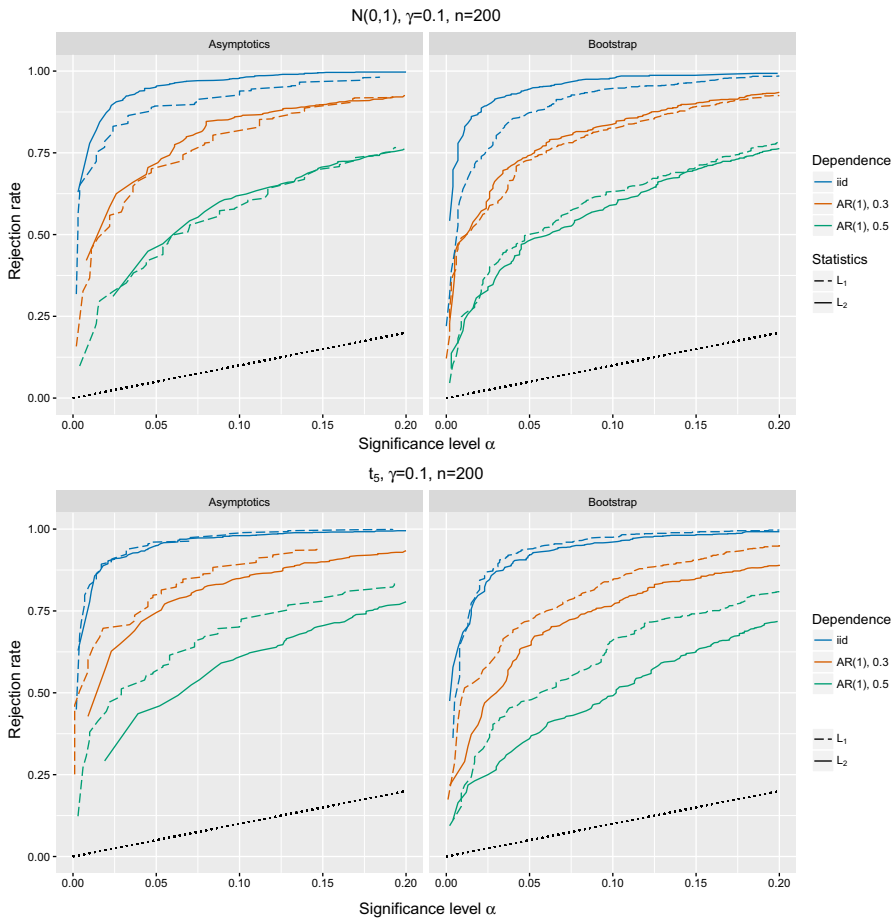


Fig. 3 Empirical (adjusted) size-power plots for $\mathcal{R}_n(\psi, \gamma)$ and $\mathcal{R}_{L,K}^*(\psi, \gamma)$, where $\gamma = 0.1$

In order to ensure that a test statistic is asymptotically distribution-free under the null hypothesis, it is necessary to use a suitable estimator of variance for the underlying process of random errors. The minimal requirement for $\hat{\sigma}_n^2(M)$ would be consistency under the null hypothesis and boundedness (in probability) under the alternative. Often, the *Bartlett estimator* is used to estimate the variance

$$\hat{\sigma}_n^2(M) = \hat{R}(0) + 2 \sum_{1 \leq k \leq M} \left(1 - \frac{k}{M}\right) \hat{R}(k), \quad M < n,$$

where

$$\hat{R}(k) = \frac{1}{n} \sum_{1 \leq i \leq n-k} (Y_i - \bar{Y}_n)(Y_{i+k} - \bar{Y}_n), \quad 0 \leq k < n.$$

However, it does not always provide satisfactory results and finding a proper estimator may be troublesome. The rate of convergence is small even under the null hypothesis and $\hat{\sigma}_n^2(M)$ might go to infinity under the alternative (Horváth et al. 2008). Other similar types of estimators can be used instead, for instance Parzen kernels (Andrews 1991), but they still possess the described deficiency.

The consistency properties of the above described Bartlett estimator and of its modification are studied in Antoch et al. (1997). The authors also describe difficulties of long run variance estimation when detecting a change in the mean of a linear process in more detail. A simulation study shows that it is not easy to find a variance estimate that would work well both under null hypothesis and under alternative. Furthermore, such estimators are often very sensitive to the *choice of the window length* M . Based on simulation studies performed by Antoch et al. (1997), we decided to use a rule of thumb of $M = n/10$.

Another representative of classical change point detection procedures is a *likelihood ratio* (LR) test statistic (for overview, see Csörgő and Horváth (1997, Chapter 1))

$$\mathcal{L}_n(M) := \frac{1}{\sqrt{\hat{\sigma}_n^2(M)}} \max_{1 \leq k \leq n-1} \left\{ \left[\frac{k}{n+1} \left(1 - \frac{k}{n+1} \right) \right] \left| \sum_{i=1}^n (Y_i - \bar{Y}_n)^2 - \sum_{i=1}^k (Y_i - \bar{Y}_k)^2 - \sum_{i=k+1}^n (Y_i - \tilde{Y}_k)^2 \right| \right\}, \quad (9)$$

where $\tilde{Y}_k = 1/(n-k) \sum_{i=k+1}^n Y_i$. This particular LR statistic represents a member of the *parametric* class of methods that relies on normally distributed observations. Asymptotic distributions of the test statistics (8) and (9) can be found, e.g., in Csörgő and Horváth (1997). The corresponding critical values come from Kulperger (1990).

Now, we demonstrate the performance of our approaches—the asymptotic and bootstrap versions of the L_1 and L_2 ratio type test statistics—compared to the traditional CUSUM and LR test statistics. This comparison is conducted within a setup, where we consider AR(1) with autoregression coefficient 0.5 (in order to mimic at least intermediate dependence) and with t_5 - as well as t_3 -distributed innovations representing heavy tailed distributions having finite variance. Empirical sizes under the null hypothesis and empirical powers under the alternative ($\tau = n/2$ and $\delta = 1$) of our four detection procedures compared to the two standard ones are shown in Table 2. The number of repetitions for the simulation of the rejection rates is set to 1000. The sample size is chosen as $n = 500$, the bootstrap block length is $K = 10$, the window smoothing length for the variance estimate is $M = 50$, and $\gamma = 0.1$.

To conclude, increase in sample size ($n = 500$ versus $n = 200$ from Figs. 1 and 2) provides more precise empirical sizes and higher powers in case of all four ratio type test techniques. The L_1 approach seems to perform as the best one out of the six considered change point test statistics in the present setup. It keeps the theoretical significance level by its empirical sizes and it gives reasonably high powers. On the other hand, the CUSUM and LR test statistics yield too small empirical size (i.e., they reject more often than they should) and provide lower power compared to the ratio type

Table 2 Empirical size (under H_0) and empirical power (under H_1 with $\tau = n/2$ and $\delta = 1$) of the tests (in percentages), where the theoretical significance level is α , $M = n/10$, $K = 10$, and $\gamma = 0.1$

Innovations	Test statistic	H_0, α			H_1, α		
		10%	5%	1%	10%	5%	1%
t_5	$\mathcal{R}_n(\psi_{L_1}, \gamma)$	11.8	7.6	1.1	86.4	76.4	54.0
	$\mathcal{R}_n(\psi_{L_2}, \gamma)$	21.9	14.8	6.9	90.9	82.0	62.4
	$\mathcal{R}_{L,K}^*(\psi_{L_1}, \gamma)$	9.8	5.6	1.6	83.3	75.3	51.4
	$\mathcal{R}_{L,K}^*(\psi_{L_2}, \gamma)$	19.6	8.7	6.6	83.7	73.2	50.2
	$\mathcal{C}_n(M)$	5.7	2.4	0.3	92.7	74.6	10.7
	$\mathcal{L}_n(M)$	6.2	2.2	0.2	92.6	76.7	2.5
t_3	$\mathcal{R}_n(\psi_{L_1}, \gamma)$	9.7	4.8	1.2	83.3	67.2	45.1
	$\mathcal{R}_n(\psi_{L_2}, \gamma)$	27.6	18.7	9.6	80.9	66.0	44.8
	$\mathcal{R}_{L,K}^*(\psi_{L_1}, \gamma)$	12.7	4.9	2.7	83.2	72.3	43.6
	$\mathcal{R}_{L,K}^*(\psi_{L_2}, \gamma)$	23.9	18.6	7.7	70.1	59.5	37.4
	$\mathcal{C}_n(M)$	7.7	2.6	0.4	73.4	51.5	7.3
	$\mathcal{L}_n(M)$	6.9	2.8	0.3	73.0	55.9	1.6

The random errors are AR(1) with autoregression coefficient 0.5 of length $n = 500$ with t -distributed innovations

test procedures, especially for small significance levels (5% and 1%). Comparison of the L_1 and the L_2 methods as well as the asymptotic and the bootstrap versions ends up with the same conclusions as previously from the size-power plots with lower sample size $n = 200$: the L_1 approaches keep the theoretical significance level under the null hypothesis more firmly compared to the L_2 ones; the L_2 methods provide higher powers than the L_1 methods; the bootstrap techniques yield more precise empirical sizes; and the asymptotic versions have higher powers. The heavier tails (t_3 against t_5) give worse results in general for all test statistics.

6 Practical application

We illustrate our testing procedures on a time series that contains $n = 619$ monthly temperature anomalies for the Northern Hemisphere from March 1958 to September 2009, see Fig. 4. The data set is obtained from R package `multitaper` (version 1.0-12) named as `HadCRUTnh` and comprises a part of data sets that have been developed by the Climatic Research Unit (University of East Anglia) in conjunction with the Hadley Centre (UK Met Office).

We want to test whether or not a change in the mean of the monthly temperature anomalies for the Northern Hemisphere occurred, assuming that the mean of the monthly temperature anomalies is approximately constant in the months before and after the possible change. The asymptotic and bootstrap version of the ratio type test statistics are considered for the L_1 and L_2 score function, where the block length is $K = 15$, the number of blocks is $L = 41$ and the number of bootstrap replications

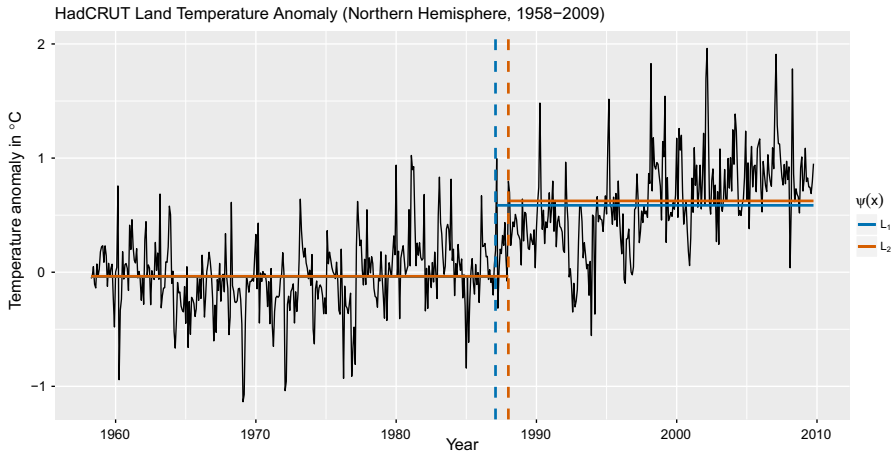


Fig. 4 Hadley Climate Research Unit Temperature anomaly (Northern Hemisphere) time series consisting of monthly observations (from March 1958 to September 2009) with estimated change point in January 1988 (L_2 -method) or in February 1987 (L_1 -method)

Table 3 Ratio type test statistics (asymptotic and bootstrap based, using L_1 and L_2 method) together with non-ratio type test statistics (CUSUM and LR test statistic) and corresponding critical values for the monthly temperature anomalies for the Northern Hemisphere for 1958–2009, considering a significance level of 5%

	Test statistic	Critical value
$\mathcal{R}_{619}(\psi_{L_1}, 0.1)$	20.29	7.38
$\mathcal{R}_{619}(\psi_{L_2}, 0.1)$	44.53	7.38
$\mathcal{R}_{41,15}^*(\psi_{L_1}, 0.1)$	20.29	6.14
$\mathcal{R}_{41,15}^*(\psi_{L_2}, 0.1)$	44.53	6.75
$\mathcal{C}_{619}(60)$	1.46	1.36
$\mathcal{L}_{619}(60)$	2.14	1.85

is $B = 10,000$ in case of the circular moving block bootstrap; and $\gamma = 0.1$. The values of the test statistics $\mathcal{R}_n(\psi, \gamma)$ and $\mathcal{R}_{L,K}^*(\psi, \gamma)$ together with the corresponding critical values are given in Table 3. All four testing procedures strongly indicate that there is a change point in the mean level, because the value of the test statistic is higher than the corresponding critical value in each case. Furthermore, the CUSUM test statistic $\mathcal{C}_n(M)$ as well as the LR test statistic $\mathcal{L}_n(M)$ reject the null hypothesis of no change.

As a by-product, we also estimated the change point as

$$\hat{\tau}_n(\psi) = \arg \max_{1 \leq i \leq n} \left| \sum_{k=1}^i \psi(Y_k - \hat{\mu}_{1n}(\psi)) \right|.$$

The L_1 and L_2 approaches provide estimated change point $\hat{\tau}_{619}(\psi_{L_1}) = 346$ months (February 1987) and $\hat{\tau}_{619}(\psi_{L_2}) = 357$ months (corresponding to January 1988), respectively. Both change points, that are distanced from each other by less than one year, are marked in Fig. 4.

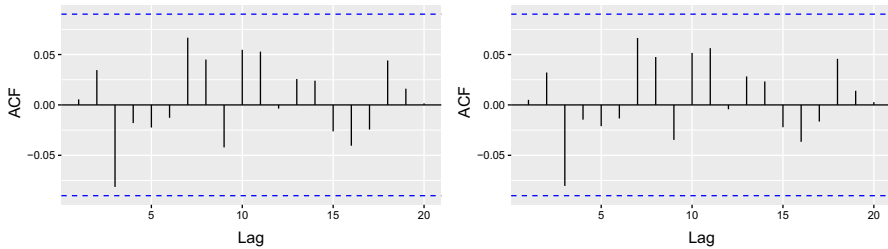


Fig. 5 Autocorrelation functions (ACFs) for the L_1 method (left) and for the L_2 method (right)

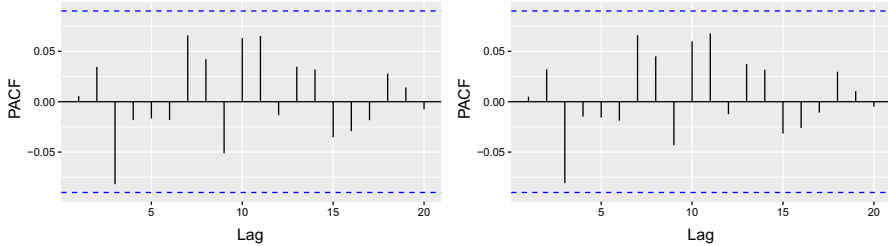


Fig. 6 Partial autocorrelation functions (PACFs) for the L_1 method (left) and for the L_2 method (right)

Finally, we investigated eligibility of the model. Let us recall the previous discussion regarding Assumptions A1–A5 on page 6. For both L_1 and L_2 methods, it suffices to check whether the fitted errors represented by the *residuals*

$$\widehat{\varepsilon}_k(\psi) := \begin{cases} Y_k - \widehat{\mu}_1 \widehat{\tau}_n(\psi)(\psi), & k \leq \widehat{\tau}_n(\psi), \\ Y_k - \widehat{\mu}_2 \widehat{\tau}_n(\psi)(\psi), & k > \widehat{\tau}_n(\psi); \end{cases}$$

come from a stationary ARMA(1,1) process. Thus, an ARMA(1,1) model is fitted to the residuals $\{\widehat{\varepsilon}_k(\psi)\}_{k=1}^n$. The ACF (autocorrelation function) and PACF (partial autocorrelation function) plots of the residuals from the fitted ARMA(1,1) model are employed (cf. Figures 5 and 6). Both ACF plots (for the L_1 method and for the L_2 method) as well as both PACF plots suggest suitability of the ARMA(1,1) model for the residuals $\{\widehat{\varepsilon}_k(\psi)\}_{k=1}^n$. We conducted the Ljung–Box test on the residuals of the fitted ARMA(1,1) models (for the L_1 method and for the L_2 method). The hypothesis that the residuals in each ARMA(1,1) model have no autocorrelation is not rejected in both cases (p values 0.559 for the L_1 method and 0.618 for the L_2 method, respectively), which suggests that the two series are stationary ARMA(1,1). The small p values (both < 0.001) provided by the augmented Dickey–Fuller (ADF) test imply that the residuals $\{\widehat{\varepsilon}_k(\psi)\}_{k=1}^n$ are stationary for the L_1 as well as for the L_2 approach. Furthermore, the L_1 approach assumes a symmetric distribution of the errors. The plot of residuals serving as the fitted errors (Fig. 7—left) and their histogram (Fig. 8—left) do not indicate violation of the symmetry assumption. The L_2 procedure requires finite moments of the errors, which cannot generally be tested (and, thus, we have to only believe in it). Zero mean of the errors in case of the L_2 method can be justified again via the plot of residuals (Fig. 7—right) and the histogram of residuals (Fig. 8—right).

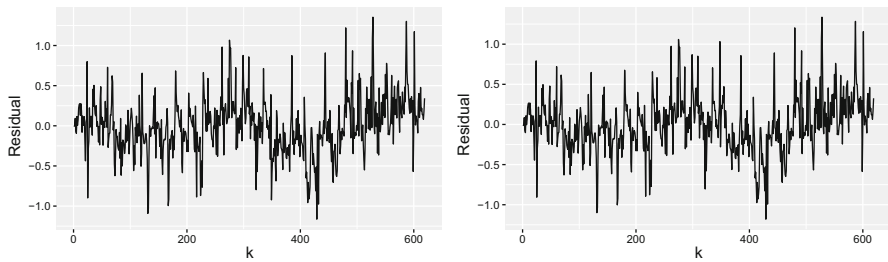


Fig. 7 Residuals $\{\hat{\varepsilon}_k(\psi)\}_{k=1}^n$ for the L_1 method (left) and for the L_2 method (right)

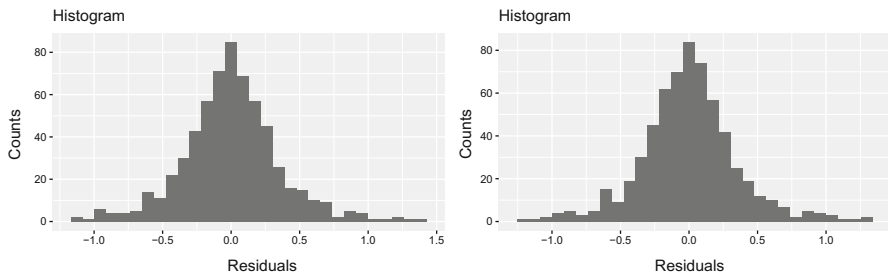


Fig. 8 Histograms of the residuals $\{\hat{\varepsilon}_k(\psi)\}_{k=1}^n$ for the L_1 method (left) and for the L_2 method (right)

7 Summary

The ratio type test statistics provide an alternative to the non-ratio type test statistics in situations, in which variance estimation is problematic. The change point detection in the *location model with at most one abrupt change in mean* is discussed in a non-parametric fashion, without assuming underlying parametric distribution. Asymptotic behavior of the ratio type test statistics is studied under the null hypothesis of no change and under the alternative of a change occurring at some unknown time point. We generalize testing procedures by assuming *weakly dependent errors* of the model together with incorporating a *general score function* in the test statistics. The circular block bootstrap method is investigated. The limit behavior of the bootstrap version of the test statistic is derived, which remains true both under the null hypothesis and under the alternative. We prove that the *block bootstrap method provides asymptotically correct critical values* for the studied ratio type test statistics in the location model with the α -mixing random errors. The simulation study illustrates that even for a relatively small time series' length, both presented approaches—based on the traditional asymptotics and on the bootstrapping—work fine, while various simulation scenarios are considered. The simulations reveal that both methods keep the significance level under the null, but the block bootstrapping seems to be more accurate. Finally, the proposed methods are applied to the real data.

Acknowledgements The authors would like to thank the Associate Editor and two anonymous referees for careful reading of the paper and for providing suggestions that improved this paper.

A Proofs

Proof of Theorem 1 The proof is analogous in several steps with the proof of Theorem 1.1 in Horváth et al. (2008). Without loss on generality, we assume that $\mu = 0$. Let

$$Z_n(t) = \frac{1}{\sqrt{n}} \sum_{1 \leq j \leq nt} \psi(\varepsilon_j) \quad \text{and} \quad \tilde{Z}_n(t) = \frac{1}{\sqrt{n}} \sum_{nt < j \leq n} \psi(\varepsilon_j).$$

By applying Theorem 1 from Doukhan (1994, Section 1.5.1) with the consequent remark justified by Herrndorf (1983) and Bulinskii (1987, 1989), we get

$$(Z_n(t), \tilde{Z}_n(t)) \xrightarrow[n \rightarrow \infty]{\mathcal{D}^2[0,1]} \sigma(\psi) (\mathcal{W}(t), \tilde{\mathcal{W}}(t)), \quad (10)$$

where $\{\mathcal{W}(t), 0 \leq t \leq 1\}$ is a standard Wiener process and $\tilde{\mathcal{W}}(t) = \mathcal{W}(1) - \mathcal{W}(t)$. Consequently, Lemma 4.3 and Lemma 4.4 by Hušková and Marušiačková (2012) together with Assumptions A1–A5 lead to

$$\sup_{1 \leq i \leq nt} \left\{ n^\kappa \sqrt{\frac{[nt]}{i([nt] - i)}} \left| \sum_{1 \leq j \leq i} \psi(Y_j - \hat{\mu}_{1,[nt]}(\psi)) - \left(\sum_{1 \leq j \leq i} \psi(\varepsilon_j) - \frac{i}{[nt]} \sum_{1 \leq j \leq nt} \psi(\varepsilon_j) \right) \right| \right\} \xrightarrow[n \rightarrow \infty]{\mathbb{P}} 0,$$

for some $\kappa > 0$, where $[a]$ denotes the integer part of $a \in \mathbb{R}$. Hence,

$$\begin{aligned} & \frac{1}{\sqrt{n}} \sup_{1 \leq i \leq nt} \left| \sum_{1 \leq j \leq i} \psi(Y_j - \hat{\mu}_{1,[nt]}(\psi)) \right| \\ &= \sup_{1 \leq i \leq nt} \left| Z_n\left(\frac{i}{n}\right) - \frac{i}{[nt]} Z_n(t) \right| + o_{\mathbb{P}}(1), \quad n \rightarrow \infty. \end{aligned}$$

Similarly, we get

$$\begin{aligned} & \frac{1}{\sqrt{n}} \sup_{nt < i \leq n} \left| \sum_{i \leq j \leq n} \psi(Y_j - \hat{\mu}_{2,[nt]}(\psi)) \right| \\ &= \sup_{nt < i \leq n} \left| \tilde{Z}_n\left(\frac{i}{n}\right) - \frac{n-i}{n-[nt]} \tilde{Z}_n(t) \right| + o_{\mathbb{P}}(1), \quad n \rightarrow \infty. \end{aligned}$$

With respect to (10), we get for all $0 < \gamma < 1/2$

$$\begin{aligned} & \left(\frac{1}{\sqrt{n}} \sup_{1 < i \leq nt} \left| \sum_{1 \leq j \leq i} \psi(Y_j - \widehat{\mu}_{1,[nt]}(\psi)) \right|, \right. \\ & \left. \frac{1}{\sqrt{n}} \sup_{nt-1 < i \leq n-1} \left| \sum_{i+1 \leq j \leq n} \psi(Y_j - \widehat{\mu}_{2,[nt]}(\psi)) \right| \right) \\ & \xrightarrow[n \rightarrow \infty]{\mathcal{D}^2[\gamma, 1-\gamma]} \sigma(\psi) \left(\sup_{0 \leq u \leq t} \left| \mathcal{W}(u) - \frac{u}{t} \mathcal{W}(t) \right|, \right. \\ & \left. \sup_{t \leq u \leq 1} \left| \widetilde{\mathcal{W}}(u) - \frac{1-u}{1-t} \widetilde{\mathcal{W}}(t) \right| \right). \end{aligned}$$

The continuous mapping theorem completes the proof

$$\mathcal{R}_n(\psi, \gamma) \xrightarrow[n \rightarrow \infty]{\mathcal{D}} \sup_{\gamma \leq t \leq 1-\gamma} \frac{\sup_{0 \leq u \leq t} |\mathcal{W}(u) - u/t \mathcal{W}(t)|}{\sup_{t \leq u \leq 1} |\widetilde{\mathcal{W}}(u) - (1-u)/(1-t) \widetilde{\mathcal{W}}(t)|}.$$

□

Proof of Theorem 2 Let $k > \tau + 1$ and $k = [\xi n]$ for some $\zeta < \xi < 1 - \gamma$. Note that $\tau = O(n)$ and $k = O(n)$ as $n \rightarrow \infty$. By the mean value theorem, we get

$$\begin{aligned} 0 &= \sum_{i=1}^k \psi(Y_i - \widehat{\mu}_{1k}(\psi)) \\ &= \sum_{i=1}^k \psi(Y_i - \mu) + \left[\sum_{i=1}^k \frac{d}{d\mu} \psi(Y_i - \mu) \Big|_{\mu=\mu^*} \right] (\widehat{\mu}_{1k}(\psi) - \mu), \end{aligned} \quad (11)$$

where μ^* lies between μ and $\widehat{\mu}_{1k}(\psi)$. The first sum in (11) can be expanded using Lemma 4.3 by Huřková and Maruřiaková (2012) and Assumptions A1–A4 as

$$\begin{aligned} \sum_{i=1}^k \psi(Y_i - \mu) &= \sum_{i=1}^k \psi(\varepsilon_i + \delta_n \mathcal{I}\{i > \tau\}) \\ &= \sum_{i=1}^k \psi(\varepsilon_i) + \sum_{i=1}^k \mathbf{E} \psi(\varepsilon_i + \delta_n \mathcal{I}\{i > \tau\}) \\ &\quad + o_{\mathbf{P}}(k^{\theta-\nu+1}) \end{aligned} \quad (12)$$

as $k \rightarrow \infty$ for any $\theta \in [-1/2, 0]$ and $\nu \in (0, \eta/(3(2 + \chi + \chi')))$. The Taylor expansion of ψ in the neighborhood of 0 with respect to Assumption A4 provides

$$\begin{aligned} \sum_{i=1}^k \mathbb{E} \psi(\varepsilon_i + \delta_n \mathcal{I}\{i > \tau\}) &= \sum_{i=\tau+1}^k \mathbb{E} \psi(\varepsilon_i + \delta_n) = k(1 - \zeta/\xi) \delta_n \lambda'(0) \\ &\quad + k(1 - \zeta/\xi) o(\delta_n), \quad k \rightarrow \infty. \end{aligned} \quad (13)$$

The second sum in (11) can be rewritten using Lemma 4.4 by Hušková and Marušíaková (2012) with respect to the Lipschitz property from Assumption A4 as

$$\begin{aligned} \sum_{i=1}^k \left. \frac{d}{d\mu} \psi(Y_i - \mu) \right|_{\mu=\mu^*} &= \sum_{i=1}^k \left. \frac{d}{dd} \mathbb{E} \psi(\varepsilon_i + \delta_n \mathcal{I}\{i > \tau\} - d) \right|_{d=0} \\ &\quad + O_{\mathbb{P}} \left(k^{1/2+(1/2+\theta)/(3+\chi)} \right) \\ &= - \sum_{i=1}^k \lambda'(\delta_n \mathcal{I}\{i > \tau\}) = -k(\lambda'(0) + O(\delta_n)) \\ &\quad + O_{\mathbb{P}} \left(k^{1/2+(1/2+\theta)/(3+\chi)} \right) \end{aligned} \quad (14)$$

as $k \rightarrow \infty$ for any $\theta \in [-1/2, 0]$. Combining (11)–(14), we end up with

$$\begin{aligned} \widehat{\mu}_{1k}(\psi) - \mu &= \frac{\sum_{i=1}^k \psi(\varepsilon_i) + k(1 - \zeta/\xi) \delta_n \lambda'(0) + k(1 - \zeta/\xi) o(\delta_n) + o_{\mathbb{P}}(k^{\theta-\nu+1})}{k(\lambda'(0) + O(\delta_n)) + O_{\mathbb{P}}(k^{1/2+(1/2+\theta)/(3+\chi)})} \\ &= \frac{1}{k\lambda'(0)} \sum_{i=1}^k \psi(\varepsilon_i) + (1 - \zeta/\xi) \delta_n \\ &\quad + o_{\mathbb{P}}(k^{\theta-\nu}), \end{aligned} \quad (15)$$

since $\delta_n = O(k^\theta)$ for some $\theta \in \left(-\frac{1}{2}, \frac{\eta}{3(2+\chi+\chi')} - \frac{1}{2}\right) \subset [-1/2, 0]$.

Consequently, applying Lemma 4.3 by Hušková and Marušíaková (2012) again, we obtain

$$\max_{1 \leq i \leq k} k^{v-\theta-1} \left| \sum_{j=1}^i (\psi(Y_j - \widehat{\mu}_{1k}(\psi)) - \psi(\varepsilon_j) - \mathbb{E} \psi(Y_j - \widehat{\mu}_{1k}(\psi))) \right| \xrightarrow[n \rightarrow \infty]{\mathbb{P}} 0. \quad (16)$$

Assumption A1 allows us to apply law of large numbers for α -mixing sequences (Chen and Wu 1989). Hence,

$$\begin{aligned}
\sum_{j=1}^i \mathbf{E} (\psi (Y_j - \widehat{\mu}_{1k}(\psi))) &= \sum_{j=1}^i \mathbf{E} \psi (\varepsilon_j + \delta_n \mathcal{I}\{j > \tau\} + \mu - \widehat{\mu}_{1k}(\psi)) \\
&= \sum_{j=1}^i \frac{1}{k} \sum_{l=1}^k \psi (\varepsilon_l + \delta_n \mathcal{I}\{j > \tau, l > \tau\} \\
&\quad + \mu - \widehat{\mu}_{1k}(\psi)) + o_{\mathbf{P}}(i), \quad k \rightarrow \infty
\end{aligned} \tag{17}$$

uniformly for $i = 1, \dots, k$. Furthermore, due to the mean value theorem and Assumption A4, we have

$$\begin{aligned}
&\sum_{j=1}^i \frac{1}{k} \sum_{l=1}^k \psi (\varepsilon_l + \delta_n \mathcal{I}\{j > \tau, l > \tau\} + \mu - \widehat{\mu}_{1k}(\psi)) \\
&= \sum_{j=1}^i \frac{1}{k} \sum_{l=1}^k \psi (\varepsilon_l) + \sum_{j=1}^i \frac{1}{k} \sum_{l=1}^k \left[\frac{d}{de} \psi (\varepsilon_l + e) \Big|_{e=e^*} \right] \\
&\quad (\mu - \widehat{\mu}_{1k}(\psi) + \delta_n \mathcal{I}\{j > \tau, l > \tau\}) \\
&= \frac{i}{k} \sum_{l=1}^k \psi (\varepsilon_l) - \frac{1}{k} \sum_{l=1}^k (\lambda'(0) + O(\delta_n)) \sum_{j=1}^i \\
&\quad (\mu - \widehat{\mu}_{1k}(\psi) + \delta_n \mathcal{I}\{j > \tau, l > \tau\}),
\end{aligned} \tag{18}$$

where e^* is between 0 and $\mu - \widehat{\mu}_{1k}(\psi) + \mathcal{I}\{j > \tau, l > \tau\}$. Plugging (15) into (18) yields

$$\begin{aligned}
&-\frac{1}{k} \sum_{l=1}^k (\lambda'(0) + O(\delta_n)) \sum_{j=1}^i (\mu - \widehat{\mu}_{1k}(\psi) + \delta_n \mathcal{I}\{j > \tau, l > \tau\}) \\
&= \lambda'(0) \left(\frac{i}{k\lambda'(0)} \sum_{s=1}^k \psi (\varepsilon_s) + i(1 - \zeta/\xi)\delta_n \right. \\
&\quad \left. - i(1 - (\tau + 1)/i)(1 - \zeta/\xi)\delta_n + i o_{\mathbf{P}}(k^{\theta-\nu}) \right) \\
&= \frac{i}{k} \sum_{s=1}^k \psi (\varepsilon_s) + \lambda'(0)\delta_n(1 - \zeta/\xi)(\tau + 1) \\
&\quad + o_{\mathbf{P}}(k^{\theta-\nu+1}), \quad k \rightarrow \infty.
\end{aligned} \tag{19}$$

Let us take into account (16) together with (17), (18), and (19). Thus, we obtain

$$\begin{aligned}
&k^{\nu-\theta-1} \left| \sum_{j=1}^{\tau+1} \psi (Y_j - \widehat{\mu}_{1k}(\psi)) \right. \\
&\quad \left. - \left(\sum_{j=1}^{\tau+1} \psi (\varepsilon_j) - \frac{\tau+1}{k} \sum_{l=1}^k \psi (\varepsilon_l) - \lambda'(0)\delta_n(1 - \zeta/\xi)(\tau + 1) \right) \right|
\end{aligned}$$

$$\leq \max_{1 \leq i \leq k} k^{\nu-\theta-1} \left| \sum_{j=1}^i \psi(Y_j - \hat{\mu}_{1k}(\psi)) - \left(\sum_{j=1}^i \psi(\varepsilon_j) - \frac{i}{k} \sum_{l=1}^k \psi(\varepsilon_l) - \lambda'(0) \delta_n (1 - \zeta/\xi)(\tau + 1) \right) \right| \xrightarrow{\mathbb{P}} 0.$$

as $k \rightarrow \infty$. Let us choose $\nu = \theta + 1/2$. Thus, $\theta \in \left(-\frac{1}{2}, \frac{\eta}{3(2+\chi+\chi')} - \frac{1}{2}\right)$ iff $\nu \in \left(0, \frac{\eta}{3(2+\chi+\chi')}\right)$. Since $\delta_n = O(k^\theta)$ as $k \rightarrow \infty$ and

$$\frac{1}{\sqrt{k}} \left| \sum_{j=1}^{\tau+1} \psi(\varepsilon_j) - \frac{\tau+1}{k} \sum_{l=1}^k \psi(\varepsilon_l) \right| = O_{\mathbb{P}}(1), \quad n \rightarrow \infty$$

according to the proof of Theorem 1 (requiring Assumption A5), we get

$$\max_{1 \leq i \leq k} \frac{1}{\sqrt{k}} \left| \sum_{j=1}^i \psi(Y_j - \hat{\mu}_{1k}(\psi)) \right| \geq \frac{1}{\sqrt{k}} \left| \sum_{j=1}^{\tau+1} \psi(Y_j - \hat{\mu}_{1k}(\psi)) \right| \xrightarrow[n \rightarrow \infty]{\mathbb{P}} \infty.$$

Note that there is no change in the means of $Y_k, \dots, Y_n$. Again from the proof of Theorem 1, we have

$$\max_{[\xi n] \leq i \leq n-1} \frac{1}{\sqrt{n}} \left| \sum_{j=i+1}^n \psi(Y_j - \hat{\mu}_{2k}(\psi)) \right| \xrightarrow[n \rightarrow \infty]{\mathcal{D}[\gamma, 1-\gamma]} \sigma(\psi) \\ \sup_{\xi \leq u \leq 1} \left| \tilde{\mathcal{W}}(u) - \frac{1-u}{1-\xi} \tilde{\mathcal{W}}(\xi) \right|,$$

which completes the proof. $\square$

Proof of Theorem 3 Let us denote $\mathbb{P}^*(\cdot) \equiv \mathbb{P}(\cdot | Y_1, \dots, Y_n)$. Moreover, let us define $\beta_n = o_{\mathbb{P}, \mathbb{P}^*}(1)$, $n \rightarrow \infty$ for some random sequence $\{\beta_n\}_{n \in \mathbb{N}}$ as follows

$$\forall \epsilon > 0, \forall \phi > 0 : \lim_{n \rightarrow \infty} \mathbb{P} [\mathbb{P}^* [|\beta_n| \geq \epsilon] \geq \phi] = 0.$$

Lemma 4.3 and Lemma 4.4 by Huřková and Maruřiaková (2012) together with the mean value theorem used in a similar way as in the proof of Theorem 2 provide

$$\begin{aligned} & \sqrt{(l-1)K+k} \left(m_{L,K}^U(l,k) - \mu \right) \\ &= \frac{1}{\lambda'(0)\sqrt{(l-1)K+k}} \left[\sum_{r=1}^{l-1} \sum_{s=1}^K \psi(\varepsilon_{U_r+s}) + \sum_{s=1}^k \psi(\varepsilon_{U_l+s}) \right] \\ &+ \delta_n \frac{\sqrt{(l-1)K+k}}{n} \left[\sum_{r=1}^{l-1} \sum_{s=1}^K \mathcal{I}\{U_r+s > \tau\} + \sum_{s=1}^k \mathcal{I}\{U_l+s > \tau\} \right] \\ &+ o_{\mathbf{P},\mathbf{P}^*}(1), \quad L \rightarrow \infty. \end{aligned}$$

Consequently, applying Lemma 4.3 by Huřková and Maruřiaková (2012) again similarly as in the proof of Theorem 2, we obtain

$$\max_{(p,q) \in \Pi_{l,k,L,K}} \left| \sum_{i=1}^{p-1} \sum_{j=1}^K \psi(Y_{U_i+j} - m_{L,K}^U(l,k)) + \sum_{j=1}^q \psi(Y_{U_p+j} - m_{L,K}^U(l,k)) \right. \quad (20)$$

$$\begin{aligned} & - \left[\sum_{i=1}^{p-1} \sum_{j=1}^K \psi(\varepsilon_{U_i+j}) + \sum_{j=1}^q \psi(\varepsilon_{U_p+j}) \right. \\ & \left. - \frac{(p-1)K+q}{(l-1)K+k} \left(\sum_{r=1}^{l-1} \sum_{s=1}^K \psi(\varepsilon_{U_r+s}) + \sum_{s=1}^k \psi(\varepsilon_{U_l+s}) \right) \right] \quad (21) \end{aligned}$$

$$- \lambda'(0)\delta_n \left\{ \sum_{i=1}^{p-1} \sum_{j=1}^K \mathcal{I}\{U_i+j > \tau\} + \sum_{j=1}^q \mathcal{I}\{U_p+j > \tau\} \right. \quad (22)$$

$$\begin{aligned} & \left. - \left(\frac{(p-1)K+q}{(l-1)K+k} \sum_{r=1}^{l-1} \sum_{s=1}^K \mathcal{I}\{U_r+s > \tau\} + \sum_{s=1}^k \mathcal{I}\{U_l+s > \tau\} \right) \right\} \Bigg| \\ &= o_{\mathbf{P},\mathbf{P}^*}(\sqrt{KL}), \quad L \rightarrow \infty. \quad (23) \end{aligned}$$

Note that (20) contains $S_{L,K}^U(p,q,l,k)$ and the expression in square brackets in (21)–(23) can be rewritten as

$$\begin{aligned} & \sum_{i=1}^{p-1} \sum_{j=1}^K (\psi(\varepsilon_{U_i+j}) - \lambda'(0)\delta_n \mathcal{I}\{U_i+j > \tau\}) \\ & + \sum_{j=1}^q (\psi(\varepsilon_{U_p+j}) - \lambda'(0)\delta_n \mathcal{I}\{U_p+j > \tau\}) \end{aligned}$$

$$\begin{aligned}
& - \frac{(p-1)K + q}{(l-1)K + k} \left(\sum_{r=1}^{l-1} \sum_{s=1}^K (\psi(\varepsilon_{U_r+s}) - \lambda'(0)\delta_n \mathcal{J}\{U_r + s > \tau\}) \right. \\
& \left. + \sum_{s=1}^k (\psi(\varepsilon_{U_l+s}) - \lambda'(0)\delta_n \mathcal{J}\{U_l + s > \tau\}) \right) =: T_{L,K}^U(p, q, l, k).
\end{aligned}$$

Furthermore, we define

$$\begin{aligned}
\tilde{T}_{L,K}^U(p, q, l, k) &:= \sum_{j=k+1}^K (\psi(\varepsilon_{U_l+j}) - \lambda'(0)\delta_n \mathcal{J}\{U_l + j > \tau\}) \mathcal{J}\{p \geq l+1\} \\
&+ \sum_{i=l+1}^{p-1} \sum_{j=1}^K (\psi(\varepsilon_{U_i+j}) - \lambda'(0)\delta_n \mathcal{J}\{U_i + j > \tau\}) \mathcal{J}\{p \geq l+2\} \\
&+ \sum_{j=1}^q (\psi(\varepsilon_{U_p+j}) - \lambda'(0)\delta_n \mathcal{J}\{U_p + j > \tau\}) \\
&- \frac{(p-l)K + q - k}{(L-l)K + K - k} \left(\sum_{s=k+1}^K (\psi(\varepsilon_{U_l+s}) - \lambda'(0)\delta_n \mathcal{J}\{U_l + s > \tau\}) \right. \\
&\left. + \sum_{r=l+1}^L \sum_{s=1}^K (\psi(\varepsilon_{U_r+s}) - \lambda'(0)\delta_n \mathcal{J}\{U_r + s > \tau\}) \right).
\end{aligned}$$

If $\{\varepsilon_i, i \in \mathbb{N}\}$ is an α -mixing sequence, then $\{\psi(\varepsilon_i), i \in \mathbb{N}\}$ is also α -mixing, but with smaller or equal mixing coefficients than $\{\varepsilon_i, i \in \mathbb{N}\}$ (Bradley 2005, Theorem 5.2). Proof of Theorem 3.6.2 by Kirch (2006) for $q(t) = 1, t \in (0, 1)$; $e(i) = \psi(\varepsilon_i)$; and $d = -\lambda'(0)\delta_n$ together with Remarks 3.5.4 and 3.5.5 from Kirch (2006) provide, conditionally on $\varepsilon_1, \dots, \varepsilon_n$,

$$\begin{aligned}
& \left(\max_{(p,q) \in \Pi_{l,k,L,K}} \frac{|T_{L,K}^U(p, q, l, k)|}{\sigma(\psi)\sqrt{LK}}, \max_{(p,q) \in \tilde{\Pi}_{l,k,L,K}} \frac{|\tilde{T}_{L,K}^U(p, q, l, k)|}{\sigma(\psi)\sqrt{LK}} \right) \\
& \xrightarrow[L \rightarrow \infty]{\mathcal{O}^2[\gamma, 1-\gamma]} \left(\sup_{0 \leq u \leq t} |\mathcal{W}(u) - u/t \mathcal{W}(t)|, \sup_{t \leq u \leq 1} |\tilde{\mathcal{W}}(u) - (1-u)/(1-t) \tilde{\mathcal{W}}(t)| \right),
\end{aligned} \tag{24}$$

in probability $\mathbf{P}$ along $\varepsilon_1, \dots, \varepsilon_n$. In contrast to Kirch (2006), we drop the assumption of random errors being a linear process (Kirch 2006, Remark 3.5.3), because this is only assumed in order to show that the original (not bootstrapped) statistic weakly converges under the null hypothesis. The assumptions of Theorem 3 and the null hypothesis (2) provide us the asymptotic distribution of $\mathcal{R}_n(\psi, \gamma)$ from Theorem 1.

By the uniform stochastic closeness (20)–(23) in a $\mathbf{P}$ -stochastic sense and (24), we get, conditionally on $Y_1, \dots, Y_n$,

$$\begin{aligned} & \frac{1}{\sqrt{LK}} \left(\max_{(p,q) \in \tilde{\Pi}_{l,k,L,K}} \left| S_{L,K}^U(p, q, l, k) \right|, \max_{(p,q) \in \tilde{\Pi}_{l,k,L,K}} \left| \tilde{S}_{L,K}^U(p, q, l, k) \right| \right) \\ & \xrightarrow[L \rightarrow \infty]{\mathcal{D}^2[\gamma, 1-\gamma]} \sigma(\psi) \left(\sup_{0 \leq u \leq t} |\mathcal{W}(u) - u/t \mathcal{W}(t)|, \right. \\ & \quad \left. \sup_{t \leq u \leq 1} |\tilde{\mathcal{W}}(u) - (1-u)/(1-t) \tilde{\mathcal{W}}(t)| \right), \end{aligned}$$

in probability $\mathbf{P}$ along $Y_1, \dots, Y_n$. Finally, the assertion of Theorem 3 is straightforward, since the considered bootstrap statistic is a continuous function of the above vector of statistics. $\square$

References

- Andrews DWK (1991) Heteroskedasticity and autocorrelation consistent covariance matrix estimation. *Econometrica* 59(3):817–858
- Andrews DWK (1993) Tests for parameter instability and structural change with unknown change point. *Econometrica* 61(4):821–856
- Andrews DWK, Ploberger W (1994) Optimal tests when a nuisance parameter is present only under the alternative. *Econometrica* 62(6):1383–1414
- Andrews DWK, Lee I, Ploberger W (1996) Optimal changepoint tests for normal linear regressions. *J Econom* 70(1):9–380
- Antoch J, Huřková M (2001) Permutation tests in change point analysis. *Stat Probab Lett* 53(1):37–46
- Antoch J, Huřková M, Prášková Z (1997) Effect of dependence on statistics for determination of change. *J Stat Plan Infer* 60(2):291–310
- Bazarova A, Berkes I, Horváth L (2015) Change point detection with stable AR(1) errors. In: Dawson D, Kulik R, Haye MO, Szyszkowicz B, Zhao Y (eds) *Asymptotic laws and methods in stochastics, fields institute communications*, vol 76. Springer, Berlin, pp 179–193
- Bradley RC (2005) Basic properties of strong mixing conditions. A survey and some open questions. *Probab Surveys* 2:107–144
- Bulinskii AV (1987) Limit theorems under weak dependence conditions. In: *Fourth international conference on probability theory and mathematical statistics*. V.N.U. Sci Press, Utrecht, pp 307–326
- Bulinskii AV (1989) On various conditions of mixing and asymptotic normality of random fields. *Soviet Math Dokl* 37(4):443–447
- Chen Z, Tian Z (2014) Ratio tests for variance change in nonparametric regression. *Statistics* 48(1):1–16
- Chen X, Wu Y (1989) Strong law for mixing sequence. *Acta Math Appl Sin* E 5(4):367–371
- Csörgő M, Horváth L (1997) *Limit theorems in change-point analysis*. Wiley, Chichester
- Doukhan P (1994) *Mixing: properties and examples*, Lecture Notes in Statistics, vol 85. Springer, New York
- Fitzenberger B (1997) The moving block bootstrap and robust inference for linear least squares and quantile regression. *J Econom* 82:235–287
- Hall P, Horowitz JL, Jing BY (1995) On blocking rules for the bootstrap with dependent data. *Biometrika* 82(3):561–574
- Herrndorf N (1983) Stationary strongly mixing sequences not satisfying the central limit theorem. *Ann Probab* 11(3):809–813
- Horváth L, Horváth Z, Huřková M (2008) Ratio tests for change point detection. In: Balakrishnan N, Peña EA, Silvapulle MJ (eds) *Beyond parametrics in interdisciplinary research: Festschrift in Honor of Professor Pranab K. Sen*, IMS Collections, Beachwood, Ohio, vol 1, pp 293–304

- Hušková M (2004) Permutation principle and bootstrap in change point analysis. *Fields Inst Commun* 44:273–291
- Hušková M (2007) Ratio type test statistics for detection of changes in time series. In: Gomes MI, Pinto Martins JA, Silva JA (eds) *Bulletin of the International Statistical Institute, Proceedings of the 56th Session, Lisboa, vol 976*, pp 3934–3937
- Hušková M, Marušiaková M (2012) *M*-Procedures for detection of changes for dependent observations. *Commun Stat B Simul* 41(7):1032–1050
- Ibragimov IA, Linnik YV (1971) Independent and stationary sequences of random variables. Wolters-Noordhoff, Groningen
- Kim JY (2000) Detection of change in persistence of a linear time series. *J Econom* 95(1):97–116
- Kim JY, Amador RB (2002) Corrigendum to detection of change in persistence of a linear time series. *J Econom* 109(2):389–392
- Kirch C (2006) Resampling methods for the change analysis of dependent data. PhD thesis, University of Cologne, Germany
- Lahiri S, Furukawa K, Lee YD (2007) A nonparametric plug-in rule for selecting optimal block lengths for block bootstrap methods. *Stat Methodol* 4(3):292–321
- Kulperger RJ (1990) On the distribution of the maximum of brownian bridges with application to regression with correlated errors. *J Stat Comput Sim* 34(2–3):97–106
- Künsch HR (1989) The jackknife and the bootstrap for general stationary observations. *Ann Stat* 17(3):1217–1241
- Leybourne SJ, Taylor AR (2006) Persistence change tests and shifting stable autoregressions. *Econ Lett* 91(1):44–49
- Madurkayová B (2011) Ratio type statistics for detection of changes in mean. *Acta Univ Carolinae Math Phys* 52(1):47–58
- Perron P (2006) Dealing with structural breaks. In: Hassani H, Mills TC, Patterson K (eds) *Palgrave handbook of econometrics, volume 1: econometric theory*. Palgrave Macmillan, Basingstoke, pp 278–352
- Peštová B, Pešta M (2015) Testing structural changes in panel data with small fixed panel size and bootstrap. *Metrika* 78(6):665–689
- Peštová B, Pešta M (2016) Erratum to: Testing structural changes in panel data with small fixed panel size and bootstrap. *Metrika* 79(2):237–238
- Politis DN, Romano JP (1992) A circular block-resampling procedure for stationary data. In: LePage R, Billard L (eds) *Exploring the limits of bootstrap*. Wiley, New York, pp 263–270
- Politis DN, White H (2004) Automatic block-length selection for the dependent bootstrap. *Econom Rev* 23:53–70
- Rosenblatt M (1971) *Markov processes: structure and asymptotic behavior*. Springer, Berlin
- Wang D, Guo P, Xia Z (2016) Detection and estimation of structural change in heavy-tailed sequence. *Commun Stat A Theor*. <https://doi.org/10.1080/03610926.1006780>
- Zhao W, Tian Z, Xia Z (2010) Ratio test for variance change point in linear process with long memory. *Stat Pap* 51:397–407
- Zhao W, Xia Z, Tian Z (2011) Ratio test to detect change in the variance of linear process. *Statistics* 45:189–198